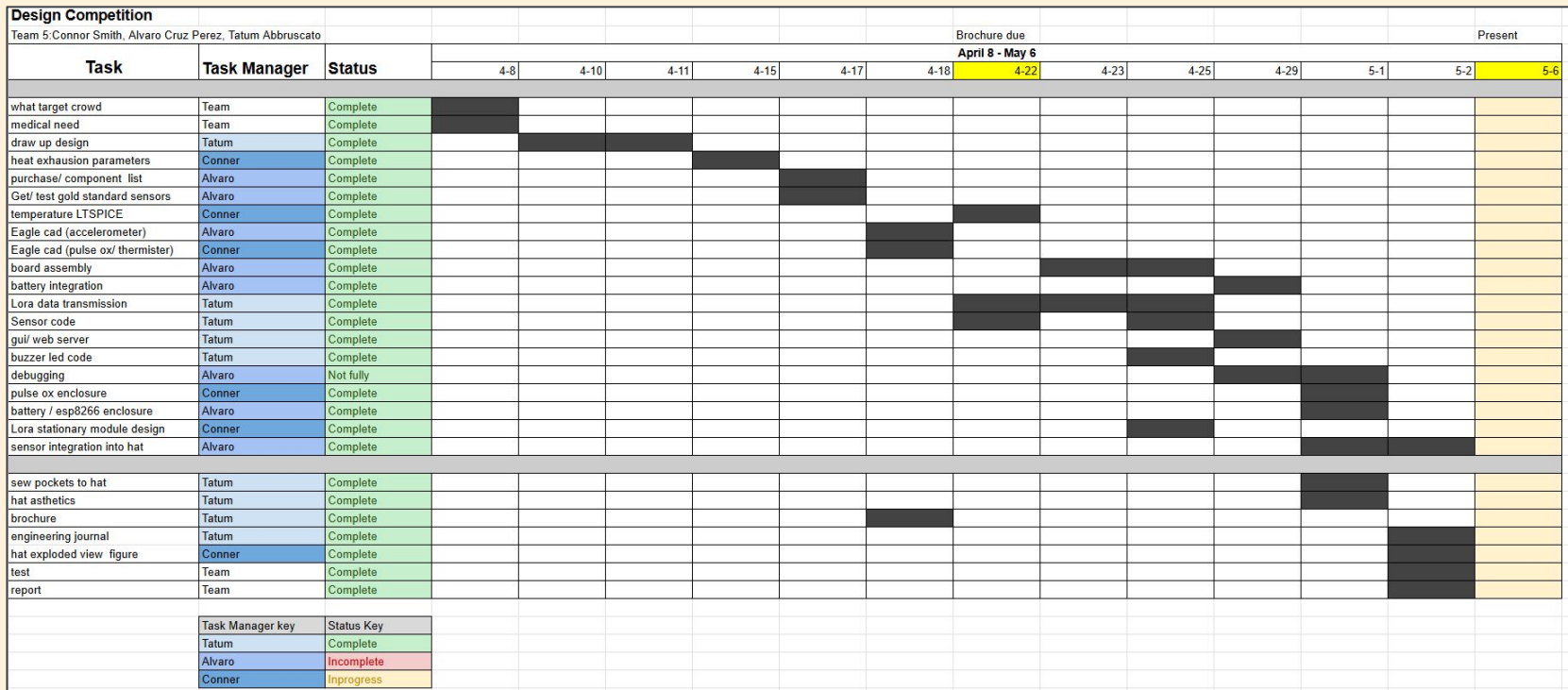




TRACK YOUR TREK

Team 5: Alvaro Cruz Perez, Tatum Abbruscato, Conner Smith

GANTT CHART



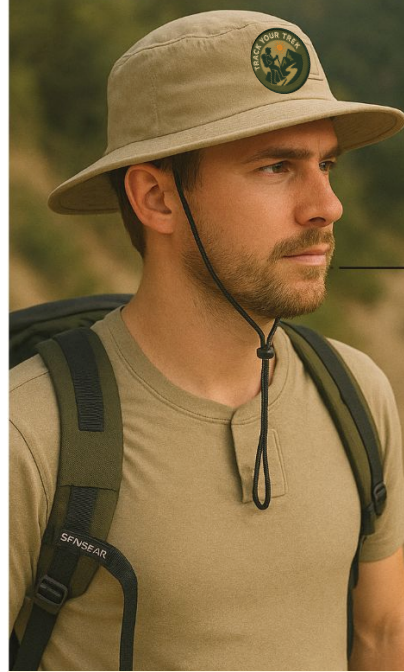
CLINICAL NEED

Increase awareness of excessive heat exposure in the hiking population by continuously monitoring environmental conditions and physiological indicators to provide clear alerts and enable early intervention and detection with the ultimate goal of reducing heat related illness.

- The National Parks System had a record 331.9 million visitors in 2024.
- According to data from the National Park Service, the 25 parks are likely to experience nearly five times more extremely hot days per year by 2050 compared to the recent past (1979-2012).
- Grand Canyon EMS responded to 474 nonfatal and 6 fatal HRI cases
- Another study found that the incidence of HRI was significantly associated with maximum daily temperature and minimum relative humidity.
- The same study estimated that HRI will increase 29%-137% over 2004-2009 levels through 2100, assuming no change in visitation.



DESIGN



HEAT EXHAUSTION PARAMETERS

- **Skin temperature** (sensed by thermistor)
- **Fainting** (sensed by accelerometer)
- **Abnormal heart rate with minimal activity** (sensed by pulse ox and accelerometer)

Condition	Core Body Temperature
Normal body temperature	97–99°F / 36.1–37.2°C
Mild hyperthermia	99–100.9°F / 37.2–38.3°C
Heat exhaustion	104°F / 40°C
Heatstroke	104°F / 40°C or higher

Condition	% Difference From Avg. BPM
Light Activity	+46.7%
Moderate Exercise	+86.7%
Vigorous Exercise	+126.7%
Heat Exhaustion	+53.3%
Heatstroke	+120%

CODE HIGHLIGHTS

Transmitting

```
// === LoRa AT Config ===
Serial.println("AT+BAND=915000000");
delay(100);
Serial.println("AT+ADDRESS=101");
delay(100);
Serial.println("AT+NETWORKID=5");
delay(100);

// === Construct LoRa message ===
String message = "x=" + String(x_g, 4) +
    ",y=" + String(y_g, 4) +
    ",z=" + String(z_g, 4) +
    ",TempF=" + String(temperatureF, 1);

if (fingerDetected) {
    message += ",BPM=" + String(beatsPerMinute, 1) +
        ",AvgBPM=" + String(avgBPM, 1);
} else {
    message += ",No contact with sensor detected";
}

int msgLength = message.length();
Serial.println("AT+SEND=102," + String(msgLength) + "," + message);
else {
    Serial.println("Failed to read accelerometer data.");
}

lastSendTime = millis();
```

Receiving

```
// Set LoRa parameters for US region
Serial.println("AT+BAND=915000000"); // Set frequency to 915 MHz
delay(100);

Serial.println("AT+ADDRESS=102"); // Set this device's address (must be the same as sender's address)
delay(100);

Serial.println("AT+NETWORKID=5"); // Set shared network ID (must match sender's network ID)
delay(100);

// LoRa module needs to be in receive mode
Serial.println("AT+RXMODE=1"); // Put LoRa module in receive mode
delay(100);
}

void loop() {
    // Check if there is any incoming message
    Serial.println("AT+RCV"); // Check for new data in receive buffer
    delay(1000);

    // If data is received, it will be printed here
    if (Serial.available()) {
        String receivedMessage = Serial.readString(); // Read the received data
        Serial.print("Received: ");
        Serial.println(receivedMessage); // Print the received message
    }
```

Display Gui

```
# === Temperature Warning Function ===
def update_warning(temp_f):
    try:
        temp_value = float(temp_f)
        if temp_value > 103:
            warning = " - Hiker may need rescue!"
            color = "red"
            text_color = "white"
        elif temp_value > 101:
            warning = " - Warning: High skin temp"
            color = "orange"
            text_color = "white"
        elif temp_value > 98:
            warning = " - Mild caution: Elevated skin temp"
            color = "yellow"
            text_color = "black"
        else:
            warning = ""
            color = "#f5deb3"
            text_color = "black"
```



PSEUDOCODE

START PROGRAM

INITIALIZE serial communication
WAIT for LoRa module to power up

INITIALIZE I2C communication on specific GPIO pins

```
// === Initialize MAX30105 Heart Rate Sensor ===  
IF MAX30105 sensor not detected THEN  
    DISPLAY error and STOP execution  
ELSE  
    CONFIGURE MAX30105 with desired LED settings  
    DISPLAY initialization success message
```

```
// === Initialize LIS2DH Accelerometer ===  
SEND configuration bytes to enable accelerometer with  $\pm 2g$  range
```

WAIT briefly for sensors to stabilize

```
// === Configure LoRa Module ===  
SET LoRa band, address, and network ID via AT commands
```

DISPLAY ready message for user to place finger on sensor

MAIN LOOP:

READ infrared (IR) value from heart rate sensor
DETERMINE if finger is detected (IR value above threshold)

IF finger detected AND a heartbeat is detected THEN
 CALCULATE time difference from last beat
 COMPUTE instantaneous BPM from time difference

IF BPM is within valid range (40–180) THEN
 STORE BPM in rolling buffer
 UPDATE buffer index
 UPDATE number of samples stored (if < window size)

COMPUTE average BPM from rolling buffer

IF 5 seconds have passed SINCE last data send:
 READ raw x, y, z values from accelerometer
 CONVERT raw values to g-units

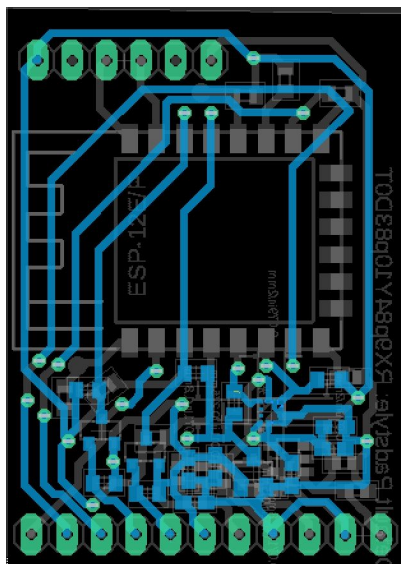
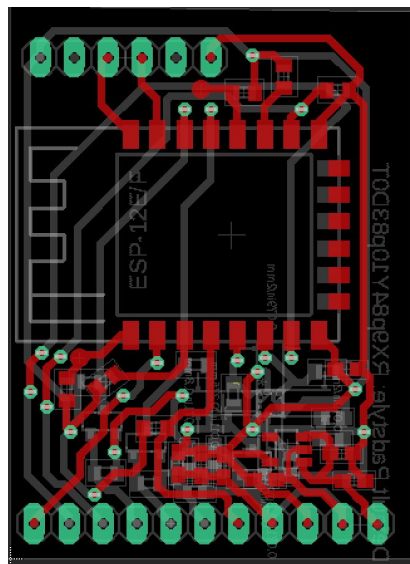
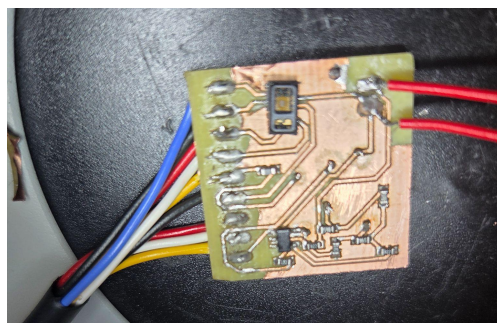
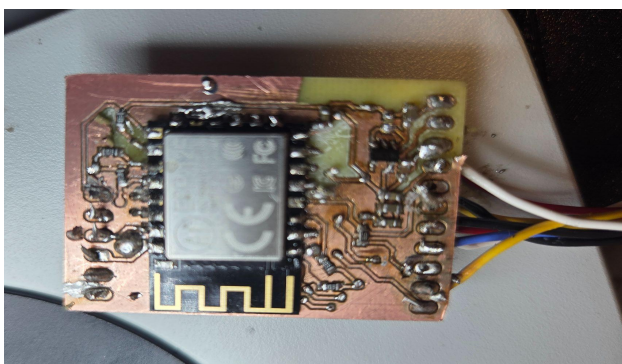
READ temperature from MAX30105 sensor
CONVERT temperature to Fahrenheit

CONSTRUCT a message string containing:
 - x, y, z acceleration
 - temperature in F
 - (optionally) current and average BPM if finger detected

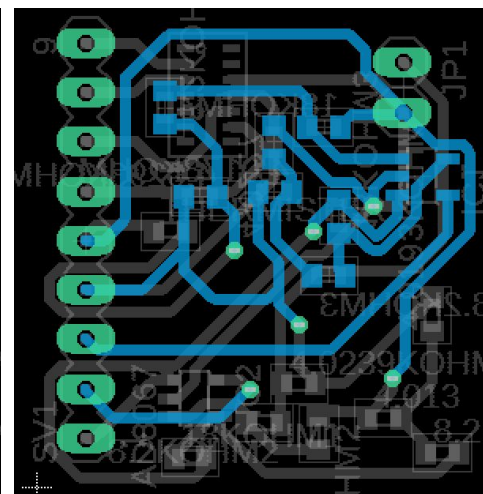
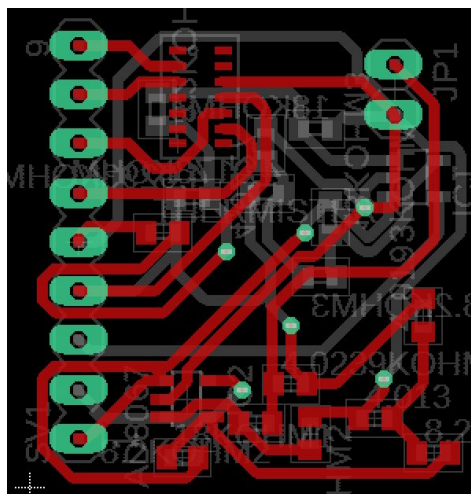
COMPUTE message length
SEND message via LoRa using AT+SEND command

UPDATE last send time

WAIT 10 milliseconds before repeating loop

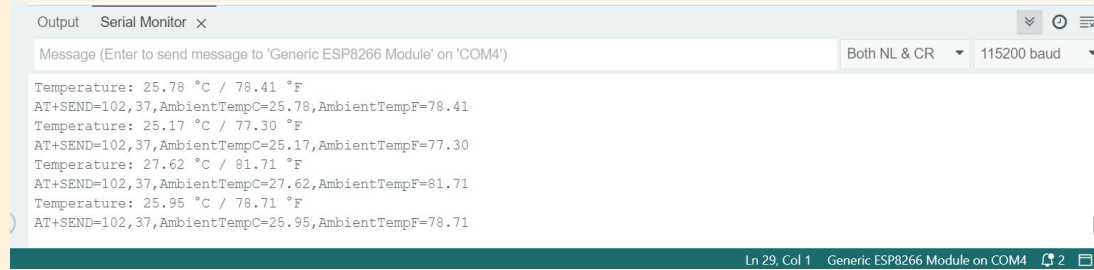


Accelerometer & ESP Board



Pulse Ox and Thermistor Boards

DEBUGGING PROCESS



The screenshot shows a Serial Monitor window with the title 'Output Serial Monitor x'. It displays a series of temperature readings in both Celsius and Fahrenheit, along with AT commands and sensor data. The status bar at the bottom indicates 'Ln 29, Col 1' and 'Generic ESP8266 Module on COM4'.

```
Message (Enter to send message to 'Generic ESP8266 Module' on 'COM4')
Both NL & CR 115200 baud

Temperature: 25.78 °C / 78.41 °F
AT+SEND=102,37,AmbientTempC=25.78,AmbientTempF=78.41
Temperature: 25.17 °C / 77.30 °F
AT+SEND=102,37,AmbientTempC=25.17,AmbientTempF=77.30
Temperature: 27.62 °C / 81.71 °F
AT+SEND=102,37,AmbientTempC=27.62,AmbientTempF=81.71
Temperature: 25.95 °C / 78.71 °F
AT+SEND=102,37,AmbientTempC=25.95,AmbientTempF=78.71
```

Problem

Voltage output of thermistors was not within predicted values



Solution

Calibrate voltage to accommodate for error and adjust code



Next Time

Pay more attention when designing layout of wheatstone bridges

Shorts in circuit



Be more careful with tracks and double check after each solder attempt, also check components (op amps) for viability frequently



Gather spare parts so it is easier to replace them in case of emergency

CLINICAL TRIAL

Here we present the a comparison of gold standard measurements and our recorded results

DEVICES / CATEGORIES	GOLD STANDARD	BREADBOARD	Error
HEART RATE	RESTING: 96, 97, ELEVATED:150.8 (in BPM)	107.5,103.6 147	9.38%
BODY TEMPERATURE	97.3 ,97.3, 97.6 (in deg F)	89.8, 90.9, 89.4	7.56%
MOVEMENT	NA	NA	NA

DEVICES / CATEGORIES	GOLD STANDARD	Custom Board	Error
OUTSIDE TEMP	114.1, 117, 117.5	79.7,79.5, 80.1	45.4%

ESTIMATED POWER CONSUMPTION

Pulse ox:

3.3 volts

Awake current:1.4 mA

Sleep current:1.3 mA

Accelerometer:

3.3volts

Awake current: 1.5 mA

Sleep current:1.2 mA

Temperature sensor:

3.3volts

Awake current 1.5mA

Sleep current 1 mA

Total Power Consumption:

$$3.3(1.4+1.5+1.5) = 14.52 \text{ W/hr \{Awake\}}$$

$$3.3(1.3+1.2+1.5) = 13.2 \text{ W/hr \{Asleep\}}$$

$$6(14.52)+6(13.2) = 166.32 \text{ W over a 12 hour usage cycle}$$

Sleep function: Sleep for 20 seconds, awake for 20

REFERENCES



- Ferreira, A. E., Ortiz, F. M., Almeida, T. T. d., & Costa, L. H. M. K. (2020). A visitor assistance system based on LoRa for nature forest parks. *Electronics*, 9(4), 696. <https://doi.org/10.3390/electronics9040696>
- Lin, C.-T., Ko, L.-W., Chang, M.-H., Duann, J.-R., Chen, J.-Y., Su, T.-P., Jung, T.-P., & Liang, S.-F. (2022). *Non-contact monitoring of cardiovascular responses during exercise by multimodal sensor fusion*. *Frontiers in Physiology*, 13, 974015. <https://doi.org/10.3389/fphys.2022.974015>
- Alam, M. M., Rahman, M. A., Farid, M. S., Islam, M. S., & Kim, J.-M. (2022). Wearable sensors for remote health monitoring. *Sensors*, 22(24), 9810. <https://doi.org/10.3390/s22249810>
- Hubbard, R. W., Szlyk, P. C., Armstrong, L. E., & Gaffin, S. L. (1996). Human tolerance to heat strain during exercise: Influence of age, gender, and fitness. *Aviation, Space, and Environmental Medicine*, 67(5), 442–447. <https://pubmed.ncbi.nlm.nih.gov/8588795/>
- Miners, A. L., & Shaw, M. T. (2022). Prevention of exertional heat stroke: Translating evidence into practice. *British Journal of Sports Medicine*, 56(8), 446–447. <https://bjsm.bmj.com/content/56/8/446>
- Centers for Disease Control and Prevention. (2007). Exertional heat illness among high school athletes—United States, 2005–2006. *Morbidity and Mortality Weekly Report*, 56(35), 896–898. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1978469/>
- Mayo Clinic. (n.d.). *Heat exhaustion*. Mayo Foundation for Medical Education and Research. Retrieved May 6, 2025, from <https://www.mayoclinic.org/diseases-conditions/heat-exhaustion/symptoms-causes/syc-20373250>
- M. Schoeffler. (2020, December 19). *LoRa sender & receiver communication with REYAX RYLR896 breakout boards / RYLR890 modules*. mSchoeffler. <https://mschoeffler.com/2020/12/19/lora-sender-receiver-communication-with-reyax-rylr896-breakout-boards-rylr890-modules/>
- Analog Devices Inc. (n.d.). *MAX30205 human body temperature sensor* [Product page]. Digi-Key Electronics. <https://www.digikey.com/en/products/detail/analog-devices-inc-maxim-integrated/max30205m1a-t/5880761>
- JST Sales America Inc. (n.d.). *S6B-PH-K-S (LF)(SN)* [Connector product page]. Digi-Key Electronics. <https://www.digikey.com/en/products/detail/jst-sales-america-inc/S6B-PH-K-S/926630>
- JST Sales America Inc. (n.d.). *S2B-PH-K-S (LF)(SN)* [Connector product page]. Digi-Key Electronics. <https://www.digikey.com/en/products/detail/jst-sales-america-inc/S2B-PH-K-S/926626>
- Maxim Integrated. (n.d.). *MAX30102: High-sensitivity pulse oximeter and heart-rate sensor for wearable health* [PDF datasheet]. Digi-Key Electronics. <https://www.digikey.com/htmldatasheets/production/1962678/0/0/1/max30102efd.pdf>